

Complete Cattle Rearing Guide

From planning and housing to feeding, breeding, health and a 52-week activity calendar

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Note: figures are general guidelines. Adjust for your climate, breed and local veterinary advice. Vaccine programmes and legal requirements differ by country and region.

1. Planning and choosing your enterprise

Success in cattle farming begins before the first animal arrives. Decide what you are producing, how many animals you can feed all year, and how you will sell.

Enterprise	Product	Typical features
Beef (cow-calf)	Weaned calves or store cattle	Keep breeding cows; sell calves at 6-9 months. Low labour, needs more land.
Beef (finishing/fattening)	Slaughter cattle	Buy young animals, feed well 3-8 months, sell. Fast turnover, needs good feed supply and capital.
Dairy	Milk (and surplus calves)	Daily milking, high labour, steady income, higher feed and health demands.
Dual purpose / mixed	Milk, meat, draught, manure	Hardy, flexible, common on smallholdings.

Key planning steps

- Assess land, rainfall, water source, road access and distance to markets.
- Calculate stocking rate: how many animals your pasture and fodder can support in the **worst** month (dry season), not the best.
- Start small (5-20 animals), learn, then expand. Most failures come from over-stocking and under-feeding.
- Prepare a simple business plan covering startup costs, running costs, expected sales and a cash buffer of 3-6 months.
- Identify a veterinarian or animal health officer and a buyer or dairy cooperative before you purchase animals.

Stocking rate guide

One Tropical Livestock Unit (TLU) is about a 250 kg animal. As a rough guide, natural tropical pasture supports 1 adult animal per 1-4 hectares depending on rainfall and soil (wetter and improved pasture supports far more; dry rangeland far less). Improved pasture with fodder crops can support 3-10 animals per hectare in intensive systems. Always confirm with your local extension officer.

2. Land, pasture and fencing

Pasture management

- **Good grasses:** Guinea grass, Rhodes grass, Brachiaria (Congo signal), Napier/elephant grass (cut and carry), Star grass. Mix with legumes (Stylosanthes, Centrosema, Desmodium, Lablab) for protein and to fix nitrogen.
- **Rotational grazing:** Divide land into 4-8 paddocks. Graze each for 3-7 days, then rest 28-45 days in the rains (longer in the dry season). This raises yield, breaks parasite cycles and prevents overgrazing.
- **Do not graze too short:** Move animals when grass is about 10-15 cm remaining, so plants recover quickly.
- **Fertility:** Spread manure, and apply fertiliser based on soil test where possible. Lime acid soils.
- **Weeds and bush:** Control bush encroachment and poisonous plants before stocking.
- **Fire management:** Cut firebreaks (3-5 m wide) before the dry season.

Fodder crops and conservation

Feed	Notes
Napier grass	High yield; cut every 45-60 days at 1-1.5 m height. Plant cuttings at onset of rains.
Hay	Cut at early flowering, dry to about 15% moisture, store off the ground under cover.
Silage	Maize, sorghum or Napier; chop, compact tightly, seal airtight for 6-8 weeks fermentation.
Crop residues	Maize stover, rice straw, groundnut haulms; low protein, so supplement (urea-molasses treatment can help).
Fodder trees	Leucaena, Gliricidia, Calliandra, Moringa give protein in the dry season (limit Leucaena to 30% of diet).

Fencing

- **Perimeter:** barbed wire (5-7 strands), electric fence, or live fence with thorny/hedging trees.
- **Paddock divisions:** electric fencing is cheap, flexible and effective.
- **Inspect weekly;** a broken fence is the most common cause of escape, crop damage and disease spread.

3. Housing and handling facilities

Class of animal	Space (covered)	Space (open yard)
Adult cow/bull	3.5-5 sq m	10-15 sq m
Heifer/steer	2.5-3.5 sq m	8-10 sq m
Calf (up to 6 months)	1.5-2 sq m (individual pen ~1.2 x 1.5 m)	3-4 sq m
Maternity/calving pen	12-16 sq m per cow	-

Design principles

- **Orientation:** Long axis east-west so the sun does not shine into the house; open sides for ventilation in hot climates.
- **Roof:** Roof height 3-4 m at the eaves; iron sheets with an overhang; insulation or a high ridge to reduce heat.
- **Floor:** Sloped 2-3% for drainage; concrete with grooves, or compacted earth for low-cost systems. Avoid slippery finishes.
- **Feed and water troughs:** Trough space of 60-75 cm per adult animal; raise troughs to keep feed clean.
- **Manure:** Plan a manure heap or biogas digester away from the water source. Manure is a valuable fertiliser.
- **Bedding:** Dry straw, sawdust or rice husks, changed frequently to reduce mastitis and foot problems.

Handling facilities

- **Crush or race:** A narrow race (about 70-75 cm wide) leading to a crush with a head-catch for vaccinations, treatment, pregnancy checks and tagging.
- **Holding pen and loading ramp:** Curved races reduce stress. Solid sides help calm animals.
- **Spray race or dip tank:** For tick control; hand-spray equipment for small herds.
- **Isolation pen:** Separate area for sick animals and new arrivals.

4. Water

Animal	Approximate daily water need
Calf (1-3 months)	5-10 litres
Growing animal (200 kg)	25-35 litres
Dry cow / beef cow	40-60 litres (up to 80 in extreme heat)
Lactating dairy cow	70-120 litres depending on yield (about 4 litres water per litre milk plus maintenance)

- Water must be **clean, fresh and always available**. Restricting water reduces feed intake and milk yield within hours.
- Sources: borehole, well, river (protected access), dam, harvested rainwater, piped supply. Limit river access points to prevent bank erosion and contamination.
- Place troughs in shade if possible and clean weekly. Provide multiple troughs so weaker animals can drink.
- Test water annually where salinity, nitrate or contamination is suspected.

5. Breeds and selecting animals

Type	Examples	Strengths	Watch-outs
Indigenous / tropical	Zebu types, N'Dama, West African Shorthorn, Sanga, Boran, Ankole	Heat, tick and disease tolerant; low input; good foragers	Lower milk and growth rates
Exotic dairy	Holstein-Friesian, Jersey, Ayrshire	High milk yield	Heat stress, tick-borne disease, high feed and care needs
Exotic beef	Hereford, Angus, Charolais, Brahman	Fast growth, good carcass	Need good nutrition; Brahman best for heat
Crossbreeds	Friesian x Zebu, Sahiwal crosses, Brahman crosses	Combines hardiness with productivity	Requires structured breeding plan

What to check when buying

- Buy from a reputable source with health and vaccination records; check age via teeth and records.
- Look for bright eyes, shiny coat, good body condition, sound legs and hooves, well-shaped udder (dairy), no discharges or coughing.
- Quarantine new arrivals for at least 2-4 weeks; test for brucellosis and TB where required.
- For bulls: choose sound conformation, good libido, breeding soundness exam, and a healthy scrotum; use unrelated bulls to avoid inbreeding.

6. Feeding and nutrition

Cattle are ruminants; the rumen microbes need a steady supply of fibre, energy and protein. Feed the rumen, then the animal.

Basic rules

- **Dry matter intake:** about 2.5-3% of body weight per day (a 400 kg cow eats roughly 10-12 kg of dry matter).
- Forage should make up at least 60% of the diet for most classes; high-concentrate diets risk acidosis.
- Change diets gradually over 7-14 days.
- Provide free-choice mineral lick and clean salt (30-50 g salt per animal per day).
- Feed at consistent times; give roughage before concentrates.

Class	Typical daily ration (400-450 kg cow equivalent)
Dry cow (maintenance)	Good pasture or 8-10 kg hay/stover plus 1-2 kg supplement, minerals
Late pregnancy (last 6-8 weeks)	Increase energy: 1-2 kg extra concentrate; adequate protein and minerals; avoid over-fat cows
Lactating dairy cow	Quality forage (green fodder 30-40 kg or equivalent) plus about 1 kg concentrate per 2-3 litres milk above the first 5-8 litres
Beef suckler cow	Pasture in the wet season; hay/crop residue plus protein supplement in the dry season
Growing/finishing steer	Good forage plus 1-3 kg concentrate/day; protein about 12-14% in the ration; target 0.7-1.2 kg gain/day
Breeding bull	Maintain good condition (BCS 3-3.5 out of 5); extra protein, minerals and energy before breeding

Common supplements

Supplement	Use
Cottonseed cake, groundnut cake, soybean meal, sunflower cake	Protein sources
Maize bran, wheat bran, rice bran, cassava chips	Energy sources
Molasses (2-4 kg/day max for adults)	Energy, palatability; mix with urea in treated feed
Urea (never more than about 1% of concentrate, and never feed neat)	Non-protein nitrogen; must be mixed and introduced gradually, risk of poisoning
Mineral premix, bone meal, limestone	Calcium, phosphorus, trace elements

Body Condition Scoring (BCS, scale 1 thin to 5 fat)

Score the herd regularly. Aim for BCS 3-3.5 at calving and breeding. Thin cows (BCS 2 or below) breed late or not at all; over-fat cows (4.5+) have calving difficulty and metabolic disease.

Feeding through the seasons

- **Rains:** Abundant green feed. Watch for bloat on lush legumes, and mineral imbalance (grass tetany). Conserve surplus as hay/silage.

- **Dry season:** Rely on stored fodder, crop residues, fodder trees and supplements. Calculate needs in advance: an adult needs about 10 kg dry matter per day, so a 150-day dry season for 10 cows needs roughly 15 tonnes of dry matter.

7. Breeding and reproduction

Parameter	Typical value
Heat (oestrus) cycle	21 days (18-24)
Duration of standing heat	12-18 hours
Gestation length	About 283 days (roughly 9 months and 1 week)
First breeding of heifers	When 60-65% of mature weight, usually 15-24 months depending on breed and feeding (earlier for well-fed exotic, later for indigenous)
Return to breeding after calving	Rebreed 60-90 days after calving for a 12-month calving interval
Bull to cow ratio	1 mature bull to 25-30 cows (1:15-20 for young bulls)
Pregnancy check	From 40-60 days by rectal palpation or ultrasound

Detecting heat

- **Standing to be mounted** is the sure sign. Other signs: restlessness, mounting others, bellowing, swollen vulva with clear mucus, reduced feed and milk, tail-head rubbed.
- Observe at least twice daily (early morning and evening) for 30 minutes. Record dates in a heat calendar.
- Tail paint or heat-mount detectors help on small farms.

Natural service vs artificial insemination (AI)

- **AI rule of thumb:** Cow seen in standing heat in the morning is inseminated in the evening; heat seen in the evening is inseminated next morning (about 12 hours after onset).
- AI gives access to superior genetics, avoids bull-borne diseases, and removes the cost of keeping a bull.
- If AI fails after 2-3 services, get the cow checked by a vet; consider a bull as clean-up.

Calving

- **Signs approaching:** Udder fills, vulva swells and relaxes, tail-head ligaments slacken, cow isolates herself; calving in 12-24 hours.
- Move cow to a clean, dry maternity pen 1-2 weeks before calving. Check every 2-3 hours, at night as well.
- **Normal birth:** Water bag then front feet (soles down) with nose resting on them. Calf is normally out within 1-2 hours of active straining.
- **Call a vet if:** hard straining for more than 1 hour with no progress, one leg or tail only visible, feet soles pointing up (breech) or head not visible, or cow is exhausted.
- Afterbirth should drop within 12 hours; if not expelled by 24 hours, call the vet - do not pull it.

8. Calf rearing

Calf losses occur mostly in the first weeks, and most are preventable.

When	Action
Birth	Clear nose and mouth; dip navel in 7% iodine within the first hour; dry and keep warm; ensure it stands.
First 2 hours (and within 6 at the latest)	Colostrum: feed 10% of body weight (about 3-4 litres) or let the calf suckle well; repeat in 6-12 hours. Colostrum gives all the immunity the calf will have initially.
Days 1-7	Keep the calf in a clean dry pen, tag and record, check navel and stools daily. Feed 4 litres of milk/day divided into 2 feeds.
Week 1-3	Disbud dairy calves (heat or paste) at 1-3 weeks. Castrate bull calves not for breeding at 1-3 months (rubber ring or Burdizzo).
Week 1 onwards	Offer fresh water and quality starter (18-20% protein) plus good hay from day 3-7 to develop the rumen.
Week 4-8	Milk at about 10% of body weight per day. Calf is eating 0.5-1 kg starter daily.
Week 8-10 (dairy)	Wean when the calf eats 1 kg or more of starter daily for 3 days. Wean gradually over a week.
Month 3-6	Grow on good pasture plus supplement. Dehorn if still required and not done earlier. Vaccinate as per programme. Deworm.
6-8 months (beef)	Wean beef suckler calves; feed weaners well after weaning to avoid setback.

Scours (calf diarrhoea) - the biggest killer

- Causes: infection (E. coli, rotavirus, coccidia, cryptosporidium), overfeeding, dirty pens, poor colostrum.
- Treatment: **keep feeding milk** plus oral rehydration solution (ORS) - 2 litres extra fluid per day - and call the vet. Isolate the sick calf.
- Signs of dehydration: sunken eyes, skin stays tented when pinched, cold mouth, weakness.

9. Health, vaccination and biosecurity

Daily observation is your best health tool. Healthy cattle are alert, eat well, ruminate (chew cud) and have a moist nose, bright eyes and normal dung.

Signs of illness

- Standing apart, dull eyes, not eating or not chewing cud, drooping ears.
- Fever (normal temperature 38.0-39.2 degrees C), rapid breathing, coughing, nasal discharge.
- Diarrhoea, blood in dung, bloated left flank, drop in milk.
- Limping, swollen joints, hoof lesions; swollen udder or clots in milk.

Typical vaccination and preventive programme (confirm with your vet)

Disease / task	Who / When	Notes
Anthrax and blackleg	Annually, before the rains; calves from 3-4 months	Killed or live spore vaccines; boosters per label.
Foot-and-mouth disease (FMD)	Every 6 months in endemic areas; calves from ~3-4 months, then booster after 4 weeks	Vaccine must match local strains.
Lumpy skin disease (LSD)	Annually	Also control biting insects.
Brucellosis (S19 or RB51)	Heifer calves 3-8 months, once	Never vaccinate pregnant or adult animals with S19. Zoonotic (dangerous to humans).
CBPP (contagious bovine pleuropneumonia)	Annually where present	Mandatory in many countries.
East Coast fever (ECF) / Theileriosis	Infection and treatment method where available	Ticks transmit; strict tick control is key.
Rift Valley fever, rabies, others	As advised locally	Rabies vaccinate in endemic zones.
Deworming	Every 3-4 months, or targeted by dung egg counts; more in rainy season and for calves	Rotate drug classes to avoid resistance. Include liver fluke control where snails are present.
Tick and fly control	Dip or spray every 1-2 weeks in rains, 2-4 weeks in dry season (follow product label)	Rotate acaricide classes; alternate with pour-ons.
Hoof trimming	1-2 times a year	Reduces lameness.

Common diseases and conditions

Condition	Signs	Response / prevention
Tick-borne diseases (ECF, anaplasmosis, babesiosis)	Fever, swollen lymph nodes, anaemia, red urine, depression	Vet treatment early; strict tick control; premunition/vaccination where used.
Trypanosomosis (nagana)	Anaemia, weight loss, rough coat, fever	Trypanocides; tsetse control; treated cattle; avoid infested zones.
Bloat	Swollen left side, discomfort, difficulty breathing	Emergency: walk animal, give anti-bloat (vegetable oil 250-500 ml or commercial), call vet urgently. Prevent: avoid hungry cattle onto lush legumes, feed hay first.
Mastitis	Hot swollen quarter, clots or watery milk, drop in yield	Strip milk, treat with vet-approved antibiotics, observe withdrawal times, teat dip after milking.
Milk fever (hypocalcaemia)	Around calving: weakness, cannot rise, cold ears	Emergency calcium by vet/intravenous; prevent with balanced dry-cow diet.
Ketosis	Early lactation: loss of appetite, weight loss, sweet breath	Vet, glucose/propylene glycol; feed energy well after calving.
Foot rot / lameness	Swollen hoof, foul smell, limping	Clean, trim, antibiotics; footbath (zinc or copper sulphate); dry yards.
Pneumonia	Cough, fever, nasal discharge, rapid breathing	Early antibiotics via vet; ensure ventilation, reduce stress and dust.
Worms / liver fluke	Bottle jaw, poor growth, diarrhoea, anaemia	Strategic deworming; rotate paddocks; snail control.
Ringworm, lice, mange	Skin lesions, itching	Topical treatment; isolate; improve hygiene.
Plant and chemical poisoning	Sudden illness, drooling, staggering	Remove from source, call vet urgently; keep cattle off toxic plants and treated crops.

Biosecurity

- Quarantine new animals 2-4 weeks and test if needed.
- Keep visitors, vehicles and equipment clean; foot dip at gates.
- Do not share bulls or equipment without cleaning; avoid contact with neighbouring herds through fences.
- Dispose of dead animals by burial (with lime) or burning; never let dogs or scavengers spread disease.
- Report notifiable diseases (FMD, anthrax, CBPP, etc.) immediately to the veterinary authority.
- Anthrax: never open the carcass of an animal that died suddenly.

Animal handling safety: bulls and cows with calves can be dangerous. Use crush facilities, work calmly and quietly, and never turn your back on a bull.

10. Dairy management

Lactation cycle

Phase	Length	Management
Early lactation	0-100 days	Peak milk around week 6-8. Cow is in negative energy balance - feed best forage and concentrate; watch ketosis, mastitis and heats.
Mid lactation	100-200 days	Rebreed by ~90 days; maintain body condition; steady feeding.
Late lactation	200-305 days	Milk falling; reduce concentrate; prepare to dry off.
Dry period	About 60 days	Rest and udder repair. Two phases: far-off dry (first 5 weeks) and close-up (last 3 weeks) with transition feed.

Milking hygiene (clean milk = higher price and less mastitis)

- Milk at regular times, twice daily (12-hour intervals ideally). Quiet, gentle cow handling.
- Wash hands and clean udder with clean water; dry with an individual cloth or paper.
- Strip the first squirts of each teat into a strip cup to detect mastitis.
- Milk quickly (5-8 minutes per cow), fully and gently. Dip teats in approved disinfectant after milking.
- Use clean stainless steel or food-grade plastic cans; wash with hot water and approved detergent.
- Cool milk quickly (below 4 degrees C within 2 hours ideal) or deliver to the collection centre promptly.
- Keep cows standing for 30 minutes after milking (feed at this time).
- Milk any cows with antibiotics last and discard milk for the label-stated withdrawal time.

Dairy rules of thumb

- Aim for a calving interval of 12-13 months.
- Heifers to calve at 24-30 months.
- Cull cows with chronic mastitis, poor fertility, and low yield.
- Replacement rate about 20-25% each year.

11. Beef management

- **Cow-calf:** Target one calf per cow per year, weaned at 6-8 months at 40-50% of the cow's weight. Manage calving season to match peak pasture.
- **Weaners and stockers:** After weaning, grow on pasture at 0.3-0.6 kg/day, with supplementation in dry months.
- **Finishing:** 3-6 months on good pasture plus 2-4 kg concentrate a day, or full feedlot rations. Sell at 350-500 kg live weight depending on market.
- **Castration:** Steers finish more predictably and are calmer; do it early (under 3 months).
- **Selling:** Use a weigh scale or heart-girth tape; sell direct to butchers, abattoirs or traders, or through livestock markets. Avoid fasting animals for long or overcrowding transport.

Weight estimate by heart girth (approximate): $\text{weight (kg)} = \text{girth (cm)}^2 \times \text{body length (cm)} / 10,840$. Use a weigh-band for a quick estimate.

12. Record keeping

Records turn guesswork into decisions. Keep them simple, consistent and up to date.

Record	What to write
Animal register	ID number/name, sex, birth date, dam and sire, breed, birth weight, purchase details
Breeding	Heat dates, service/AI dates, bull or semen used, pregnancy check, expected and actual calving date, calf details
Milk (dairy)	Daily yield per cow, total; fat/quality where known
Growth	Weigh at birth, weaning, 12 months, and monthly for fatteners
Health	Vaccinations, dewormings, dips, treatments, drug used, dose, withdrawal date, vet notes
Feed	Type and quantity used, cost, pasture rotation dates, rainfall
Finance	All sales and expenses, cash book, monthly summary
Deaths/culls	Date, cause, and lessons

Identify every animal with ear tags, brands, or neck chains, and keep a backup record.

13. Economics and marketing

Typical cost categories

Startup (one-off)	Running (recurring)
Animals; land purchase/lease; fencing; housing; crush; water infrastructure; milking equipment or cooling; feed store; vehicle/transport; tools	Feed and minerals; veterinary and drugs; AI/bull services; labour; water and electricity/fuel; transport; insurance; licences; repairs; interest

Where the money is usually made or lost

- **Feed is 50-70% of running costs.** Growing your own fodder and buying supplements in bulk during harvest keeps costs down.
- **Fertility drives profit:** each extra month a cow is empty costs a month of feed with no calf or milk.
- **Mortality:** a lost calf or cow wipes out several animals' profit; prevention is cheaper than cure.
- **Timing sales:** Prices are usually high before festive seasons and during the dry season when supply is short; plan finishing to hit those windows.
- **Value addition:** Yoghurt, cheese, ghee and processed meat can raise returns; sell manure or use biogas.

Simple profit check

Annual income (milk/animal sales + manure + inventory gain) minus annual cost (feed + health + labour + interest + depreciation + other) gives profit per animal. Calculate this each year and compare against your target.

Marketing options: cooperatives, abattoirs, direct to butchers, livestock markets, contract feedlots, dairy processors, restaurants and households. Agree on price, quality and payment terms before delivery.

14. Regulations, welfare and environment

- Register the farm and animals if required; keep movement permits for transport and sale.

- Follow disease control rules, notifiable disease reporting, meat inspection and milk quality standards.
- **Five freedoms of welfare:** freedom from hunger and thirst; from discomfort; from pain, injury and disease; to express normal behaviour; from fear and distress.
- Use humane methods for handling, castration (with pain relief where possible), and transport.
- Manage manure and waste to protect streams and neighbours; avoid overgrazing and soil erosion; keep grazing away from riverbanks.
- Keep safe records of drug use and honour withdrawal times before slaughter or milk sale.

15. Routine daily, weekly and monthly tasks

Frequency	Tasks
Every day	Check every animal for health, lameness and heat signs; feed and water (twice or more); milk if dairy; clean sheds and troughs; check fences and gates; check calves and cows close to calving; record milk, sick animals and heats.
Every week	Dip/spray for ticks in the rains (or as per programme); inspect fences, water points and pipes; move paddocks in rotation; clean and disinfect milking equipment; review heat and pregnancy records; check feed stocks; hoof and udder checks.
Every month	Weigh young stock and fatteners; body condition scoring; update records and cash book; check supplies of drugs and minerals; review pasture growth; deworm calves as scheduled; service pumps and equipment.
Every quarter	Deworm herd; review production and finances; pregnancy diagnosis; soil or pasture check; consult vet on health plan; sell surplus/cull.
Every year	Vaccinations; full stock take; annual budget and plan; land preparation and pasture renovation; equipment overhaul; training (attend farmer field days).

16. The 52-week year-round calendar

How to read this calendar. The plan assumes a breeding herd managed for **calving early in the year (around weeks 9-20)** just before the main rains, so calves and lactating cows benefit from lush pasture. It is built around a typical tropical rhythm: dry season (weeks 1-13 and 45-52), main rains (about weeks 14-30), a short dry spell (weeks 31-35) and minor rains (weeks 36-44). If your seasons differ, shift the whole plan so that **calving starts about 3-4 weeks before the main rains** and keep the same order of activities. Week 1 = first week of January. Mating in weeks 21-32 gives calving from week 9 of the next year (283 days). Every week also includes the daily routine in Section 15.

Wk	Season	Focus	Key activities
1	Dry season	Annual review	Count and inspect every animal; body condition score all; review last year's records (calving, milk, deaths, sales). Set targets and budget. Feed stored hay/silage plus protein supplement; check mineral licks and water supply.
2	Dry season	Repairs and safety	Repair fences, gates, crush and troughs. Clean fire breaks around pasture and sheds. Check borehole/pump; service dip or spray equipment.
3	Dry season	Stock and supplies	Weigh young stock and heifers. Order vaccines, dewormers, acaricides, minerals, iodine, ear tags, calving supplies. Confirm vet is available for calving season.
4	Dry season	Dry-season feeding	Review fodder stock; cut and carry Napier/fodder trees where available. Give late-pregnant cows extra energy and protein; separate thin cows for special feeding.
5	Dry season	Prepare calving area	Clean and disinfect maternity pens; fresh bedding. Move cows due in the next 6 weeks to a close-up group; start watching udder development. Trim hooves of cull-prone or lame animals.
6	Dry season	Pre-calving health	Vaccinate late-pregnant cows for scours (if used) 3-6 weeks before calving; check minerals (especially calcium/phosphorus balance). Tick control on schedule.
7	Dry season	Calving kit ready	Assemble kit: clean rope, gloves, lubricant, 7% iodine, warm towels, feeding bottle and tube, frozen colostrum, ear tags, notebook. Post the vet's phone number in the shed.
8	Dry season	Night watch begins	Check heavily pregnant cows every 3 hours including at night. Have a clean dry area ready; ensure electricity/torch light works. Plan land preparation for fodder crops.
9	Dry season	Calving starts	Attend births. Navel iodine within 1 hour; calf suckles colostrum within 2 hours; tag, weigh and record. Cow should pass afterbirth within 12 hours.
10	Dry season	Early calving	Continue night checks. Increase concentrate for fresh cows; keep calves in clean pens or with dams. Watch for milk fever within 48 hours of calving.
11	Dry season	Milk and colostrum	Establish milking routine for dairy cows; fresh cows on best feed. Check for retained placenta, mastitis, ketosis. Freeze extra colostrum for weak calves.
12	Dry season	Calf procedures	Disbud dairy calves (1-3 weeks). Castrate bull calves (rubber ring or Burdizzo) under vet guidance. Start offering starter feed and water to calves older than 1 week.
13	Dry season	Pre-rain preparation	Repair roofs, drains and troughs. Prepare land: slash, plough or clear fodder plots; spread manure. Buy seed and planting materials. Pre-rain vaccination plan set.
14	Main rains	Onset of rains	Plant fodder crops (Napier cuttings, maize, sorghum, legumes) at first reliable rains. Release cattle onto green pasture gradually ; give hay first to avoid scours and bloat.

Wk	Season	Focus	Key activities
15	Main rains	Vaccination campaign	Vaccinate against anthrax, blackleg, LSD and any regionally required diseases (calves from 3-4 months). Record batch numbers. Start weekly/fortnightly tick control.
16	Main rains	Rotational grazing starts	Set up paddocks and grazing calendar; graze 3-7 days, rest 30-45 days. Deworm cattle at start of rains. Check for foot rot in muddy areas.
17	Main rains	Calf growth	Calves 4-8 weeks: weigh, increase starter. Vaccinate heifer calves (3-8 months) against brucellosis. Watch for pneumonia and scours during wet weather.
18	Main rains	Mid-calving season	Continue calving and record. Check body condition of lactating cows; increase feed if losing condition. Test milk for mastitis (CMT) in dairy herds.
19	Main rains	Bull and heifer check	Breeding soundness check for bulls; feed bulls up. Select heifers to breed (by weight, at least 60-65% of mature weight). Order AI semen and straws.
20	Main rains	End of calving	Last calves born; tally survival rate and average birth weights. Cows begin returning to heat about 45-60 days after calving; start recording heat signs. Fresh-cow health checks.
21	Main rains	Breeding season begins	Turn bulls in (1 bull to 25-30 cows) or start AI. Observe heat twice daily; inseminate 12 hours after onset. Flush-feed cows and heifers (rising plane of nutrition).
22	Main rains	Breeding and heat	Continue heat detection; record every service. Tick control and weekly checks on hooves and udders. Weed fodder plots and top-dress with manure or fertilizer.
23	Main rains	Breeding and fodder	Check heifer and cow body condition; separate thin cows for extra feed. First cut of Napier/fodder crops at 45-60 days. Begin hay/silage making from surplus pasture.
24	Main rains	Health check-up	Deworm herd if rains are heavy (or use dung egg counts). Vaccinate stragglers. Hoof inspection and footbath. Review water quality and trough cleaning.
25	Main rains	Mid-year review	Compare production, calf numbers, costs, and prices with budget. Adjust stocking and feed plan. Check for signs of overgrazing and rest paddocks accordingly.
26	Main rains	Silage and hay	Make silage from maize/sorghum/Napier (chop, compact, seal for 6-8 weeks). Store hay under shelter. Record quantity conserved.
27	Main rains	Weigh and grade	Weigh weaners, steers and heifers. Identify slow growers; plan feed or cull. Continue breeding record checks; repeat services for cows that returned to heat.
28	Main rains	Breeding continues	Watch for repeat breeders (returns at 18-24 days). Vet check for cows not cycling by 90 days after calving. Keep tick and fly control (wet-season peak).
29	Main rains	Pasture management	Move animals promptly; oversow legumes in bare patches; control weeds. Clean and repair the manure heap or digester; spread compost on pasture.
30	Main rains	Wet-season peak	Watch for pneumonia, foot rot, trypanosomiasis and tick-borne diseases. Keep calves dry. Ensure minerals and salt available. Repair leaking roofs.
31	Short dry	Rains ease	Conserve more feed. Begin pregnancy diagnosis for cows served early (40-60 days after service). Keep water points clean and reliable.
32	Short dry	Close breeding	Stop breeding at week 32 (12-week season) so calving is bunched. Remove bulls from cows. Record final service list.
33	Short dry	Pregnancy diagnosis	Pregnancy-test all served cows and heifers (palpation/ultrasound). Sort empty (open) cows: cull, sell, or re-breed by AI next cycle.
34	Short dry	Hay and fodder	Bale or stack hay; cut second Napier cut. Fence off reserve pasture (standing hay) for the dry season. Estimate fodder stock versus needs.

Wk	Season	Focus	Key activities
35	Short dry	Health and hoof	Hoof trimming, dip check, and tick control adjustment. Deworm calves and weaners. Review parasite control.
36	Minor rains	Weaning time	Wean beef calves born in weeks 9-20 (now about 6-7 months); dairy calves already weaned. Feed weaners well after weaning: good pasture plus 1-2 kg supplement.
37	Minor rains	Post-weaning care	Vaccinate weaners (boosters if needed). Group by size and sex. Deworm and treat for ticks. Remove cows to separate paddock to reduce stress.
38	Minor rains	Dry off dairy cows	Dry off cows due to calve in about 60 days. Treat with dry-cow therapy as advised by vet; adjust feed to dry-cow ration.
39	Minor rains	Second silage/hay	Make hay and silage from minor-rains growth. Plant fodder trees, legumes or new pasture. Store crop residues (stover) safely.
40	Minor rains	Replacement heifers	Select replacement heifers (from best dams and fast growth). Weigh and tag. Choose culls: barren cows, poor milkers, bad temperament, chronic mastitis or lameness.
41	Minor rains	Health round	Deworm (if due), acaricide, footbath. Check body condition of pregnant cows: aim for BCS 3 by calving. Vaccinate for FMD or others if on 6-month cycle.
42	Minor rains	Sales window	Sell finished steers and culls at good prices while pasture is still good and before dry-season prices fall or feed costs rise. Deliver with permits.
43	Minor rains	Pasture rest and renovation	Rest overgrazed paddocks; oversow legumes; fertilize. Repair fences and water points before dry season.
44	Minor rains	Fodder bank	Complete fodder bank: hay, silage, stover, molasses and concentrates purchased. Calculate feed for the coming dry season (about 10 kg dry matter per adult per day).
45	Dry season	Dry season begins	Move to supplemented feeding. Treat crop residues with urea if using. Provide protein blocks or mineral licks. Watch water supplies and pumps.
46	Dry season	Feed budget	Check fodder consumption against plan; buy additional feed early while cheap. Cull additional weak or unproductive cows to save feed.
47	Dry season	Vaccination boosters	Give routine vaccinations due (FMD, LSD, CBPP as required). Update vaccination records. Deworm before dry season starts to bite.
48	Dry season	Water and shade	Check boreholes, troughs, pipes; provide shade and windbreaks. Watch for heat stress: shade, water and reduced midday handling.
49	Dry season	Festive season market	Sell finished and cull animals ahead of holidays when prices are high. Plan labour cover for the holiday period so daily care continues.
50	Dry season	Fire watch	Inspect firebreaks; keep fire-fighting tools and water ready. Guard hay stores. Continue supplementary feeding.
51	Dry season	Stock take	Full inventory of cattle, feed, drugs, equipment. Reconcile the herd register with actual animals; plan for animals due to calve in about 10-14 weeks.
52	Dry season	Year-end accounts	Close books; calculate profit per cow/litre/kg gain. Review what worked and what to improve. Draft next year's plan, budget and calendar; thank staff and repeat cycle at Week 1.

Shading: sand = dry season, green = main rains, blue = minor rains.

Calendar at a glance

Weeks	Main focus
1-8	Dry-season feeding, repairs, supplies, pre-calving preparation
9-20	Calving season; colostrum, navel care, calf health; fresh-cow care; start of rains vaccination and fodder planting
14-20	Onset of rains: planting, gradual pasture introduction, vaccination, tick control
21-32	Breeding season (bulls/AI); flush feeding; heat detection; hay and silage making
31-35	Close breeding, pregnancy diagnosis, hay, hoof and health checks
36-44	Weaning, drying off, replacement selection, culling and sales, fodder bank building
45-52	Dry-season feeding, boosters, festive sales, stock take, year-end accounts and planning

17. Troubleshooting quick reference

Problem	Likely causes	What to do
Cows not returning to heat	Poor body condition, silent heat, pregnancy, uterine infection, mineral deficiency	Check BCS, feed better, confirm pregnancy, vet exam at 60-90 days after calving.
Repeat breeders	Poor heat detection or AI timing, infection, nutrition, bull problem	Improve detection, correct timing, check for infection, change bull or semen.
Calf deaths in first weeks	Poor colostrum, dirty pens, scours, navel infection, cold/wet	Fix colostrum protocol, clean and dry pens, iodine navel, treat early.
Low milk	Poor feed or water, heat stress, mastitis, disease, late lactation	Check ration, water and shade; test for mastitis; treat causes.
Weight loss in dry season	Not enough feed or protein, worms, ticks, disease	Add supplement, deworm, tick control, cull weakest animals early.
Sudden death	Anthrax, blackleg, poisoning, bloat, lightning	Do not open carcass if anthrax suspected; report to vet authority; bury or burn.
Limping	Foot rot, wounds, overgrown hooves, laminitis, injury	Examine hoof, trim and clean, footbath, treat and improve flooring.
Bloat	Lush legumes, sudden diet change, hungry animals	Emergency measures listed in Section 9, call vet, prevent by feeding hay first.
High feed cost	Wasted fodder, poor pasture, too many low-producing animals	Improve pasture, grow fodder, ration properly, cull poor performers.

Final advice. Start small, keep records, feed for the dry season, keep animals healthy through prevention, and treat your vet and extension officer as partners. This guide is general information and not a substitute for professional veterinary and agricultural advice for your farm.